



**Introduction to**

**Internet of Things**

**Assignment-Week 3**

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions: 15**

**Total marks: 15 X 1= 15**

**QUESTION 1:**

State True or False.

Statement: “WirelessHART is the latest release of Highway Addressable Remote Transducer protocol.”

- a. True
- b. False

**Correct Answer: a. True**

**Detailed Solution:** WirelessHART is the latest release of Highway Addressable Remote Transducer protocol.

See lecture 11 (Connectivity Technologies-III) @ 01:42

**QUESTION 2:**

State True or False.

Statement: “Wired HART has a network layer.”

- a. True
- b. False

**Correct Answer: b. False**

**Detailed Solution:** Wired HART does not have a network layer.

See lecture 11 (Connectivity Technologies-III) @ 04:25



**QUESTION 3:**

State true or false:

“WirelessHART physical layer is derived from 802.15.2 protocol”

- a. False
- b. True

**Correct Answer: a. False**

**Detailed Solution:** WirelessHART physical layer is derived from 802.15.4 protocol.

See lecture 11 (Connectivity Technologies-III) @ 06:00

**QUESTION 4:**

WirelessHART operates only in \_\_\_\_\_ GHz ISM band.

- a. 3.7
- b. 4.8
- c. 4.8
- d. 2.4

**Correct Answer: d. 2.4**

**Detailed Solution:** HART operates only in the 2.4 GHz ISM band.

See lecture 11 (Connectivity Technologies-III) @ 06:00

**QUESTION 5:**

HART standard was developed from \_\_\_\_\_ smart field devices.

- a. amplified
- b. diminished
- c. isolated
- d. networked

**Correct Answer: d. networked**

**Detailed Solution:** HART standard was developed for networked smart field devices.

See lecture 11 (Connectivity Technologies-III) @ 02:36



**QUESTION 6:**

Main difference between wired and unwired versions is in the physical, data link, and \_\_\_\_\_ layers.

- a. Data link
- b. Network**
- c. Transport
- d. None of these

**Correct Answer: b. Network**

**Detailed Solution:** Main difference between wired and unwired versions is in the physical, data link, and network layers.

See lecture 11 (Connectivity Technologies-III) @ 05:05

**QUESTION 7:**

State true or false

“Collision free and deterministic communication is achieved in HART’s data link layer.”

- a. True**
- b. False

**Correct Answer: a. True**

**Detailed Solution:** Collision free and deterministic communication is achieved in HART’s data link layer.

See lecture 11 (Connectivity Technologies-III) @ 06:37

**QUESTION 8:**

Channel hopping is incorporated in which layer of HART?

- a. Data link layer**
- b. Physical layer
- c. Application layer
- d. Transport layer

**Correct Answer: a. Data link layer**

**Detailed Solution:** Channel hopping is incorporated in the data link layer of HART.

See lecture 11 (Connectivity Technologies-III) @ 08:21



**QUESTION 9:**

State True or False.

“The HART application layer is responsible for extracting commands from a message, executing it and generating responses.”

- a. True
- b. False

**Correct Answer: a. True**

**Detailed Solution:** The HART application layer is responsible for extracting commands from a message, executing it and generating responses.

See lecture 11 (Connectivity Technologies-III) @ 10:26

**QUESTION 10:**

NFC is designed for use by devices within \_\_\_\_\_ to each other.

- a. Anywhere on the globe
- b. A small building
- c. Both (a) and (b)
- d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** NFC is designed for use by devices within close proximity to each other.

See lecture 11 (Connectivity Technologies-III) @ 17:47

**QUESTION 11:**

Passive NFC devices \_\_\_\_\_ information which is \_\_\_\_\_ by other devices.

- a. contain, read
- b. read, contain
- c. contain, contain
- d. None of these

**Correct Answer: a. contain, read**

**Detailed Solution:** Passive NFC devices contain information which is read by other devices.

See lecture 11 (Connectivity Technologies-III) @ 18:34



**QUESTION 12:**

State True or False.

“NFC devices work on the principle of magnetic induction.”

- a. True
- b. False

**Correct Answer: a. True**

**Detailed Solution:** NFC devices work on the principle of magnetic induction.

See lecture 11 (Connectivity Technologies-III) @ 21:47

**QUESTION 13:**

Bluetooth technology is based on \_\_\_\_\_.

- a. HART
- b. ZigBee
- c. All of these
- d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** Bluetooth technology is based on Ad-hoc piconets.

See lecture 12 (Connectivity Technologies-IV) @ 02:30

**QUESTION 14:**

State whether the following statement is true or false.

Statement: The Link Manager Protocol in Bluetooth manages the only establishment and authentication.

- a. True
- b. False

**Correct Answer: b. False**

**Detailed Solution:** The Link Manager Protocol in Bluetooth manages the establishment, authentication, link configuration.

See Page number – 157, Chapter - 7, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)



**QUESTION 15:**

Zigbee commonly uses \_\_\_\_\_ data rate.

- a. 260 bps
- b. 260 kbps
- c. 260 Mbps
- d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** Zigbee commonly uses a 250-kbps data rate.

See Page number – 131, Chapter - 7, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

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